Critical Review of the Proposal: Conversational Elicitation of Patient Utilities Using Large Language Model Simulation

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1 Executive Summary

The proposal presents an ambitious and intellectually innovative framework for inferring patient utilities from conversational narratives using large language model (LLM) simulation and round-trip evaluation.

The project has several notable strengths:

- conceptual originality,
- strong alignment with narrative medicine,
- a reproducible open-source architecture,
- thoughtful use of simulation-based evaluation,
- strong fit to oral squamous cell carcinoma (OSCC),
- practical emphasis on low-cost infrastructure.

However, the proposal also faces substantial conceptual, methodological, psychometric, and translational challenges.

The most important concerns include:

1. ambiguity in the definition of "utility,"
2. the risk of circularity in LLM-based simulation,
3. weak grounding in psychometric theory,
4. uncertainty about identifiability,
5. possible overestimation of conversational recoverability,
6. insufficient attention to causal structure,
7. unclear validation pathways against real patient data,
8. ethical risks related to inference from narrative.

This critique argues that the project is strongest as a simulation and methodological research platform, rather than an immediate clinical utility measurement tool.

The proposal would benefit from:

- sharper theoretical definitions,
- stronger psychometric grounding,
- clearer evaluation criteria,
- explicit failure modes,
- stronger separation between simulation and inference models,
- more detailed plans for real-world validation.

2 Major Strengths

3 Conceptual Originality

The proposal introduces a genuinely novel conceptual framing:

> utility elicitation as inverse inference over conversational narratives.

This is a significant departure from traditional utility elicitation approaches.

The proposal successfully integrates ideas from:

- narrative medicine,
- latent-variable modeling,
- simulation-based inference,
- inverse reinforcement learning,
- adaptive interviewing,
- LLM-driven generative simulation.

The "round-trip" architecture is particularly compelling because it converts an otherwise subjective process into a measurable inference problem.

4 Strong Fit Between Domain and Method

OSCC is an excellent initial disease domain because:

- impairments are highly behaviorally visible,
- side effects naturally affect daily activities,
- conversational manifestations are plausible and rich,
- functional impacts are narratively expressible.

This disease choice substantially strengthens the feasibility of conversational inference.

5 Open-Source and API-Free Design

The proposed technical architecture is unusually pragmatic.

Advantages include:

- low infrastructure cost,
- reproducibility,
- transparency,
- ease of collaboration,
- long-term sustainability,
- compatibility with local models.

Avoiding mandatory API integration is an especially strong design decision for academic reproducibility.

6 Emphasis on Simulation Rather Than Immediate Clinical Deployment

The proposal wisely focuses first on:

- simulation,
- benchmarking,
- identifiability,
- methodology development.

This is scientifically appropriate given the uncertainty surrounding recoverability of utilities from language.

7 Major Conceptual Weaknesses

8 Ambiguity in the Meaning of "Utility"

The proposal uses the term "utility" somewhat inconsistently.

At different points, utilities appear to refer to:

- health-state preference values,
- attribute weights,
- symptom salience,
- emotional burden,
- quality-of-life dimensions,
- conversational attention allocation,
- personal priorities.

These are related but distinct constructs.

For example:

- A patient may frequently discuss fatigue without assigning it large utility impact.
- A patient may deeply value speech function but discuss it minimally due to adaptation.
- Emotional distress and utility decrement are not necessarily equivalent.

The proposal needs a much sharper formal definition of:

- what exactly is being inferred,
- how it maps to existing utility theory,
- whether the target construct is:
  - QALY utility,

  - preference weight,
  - latent salience,
  - experienced burden,
  - or some new construct entirely.

9 Unclear Relationship to Existing Preference-Elicitation Theory

The proposal references conventional utility methods but does not sufficiently engage with:

- utility theory,
- random utility models,
- psychometrics,
- multi-attribute utility theory,
- health-state valuation literature.

This creates a risk that the project appears conceptually detached from existing health economics methodology.

The proposal should more explicitly address:

- whether conversationally inferred utilities are intended to substitute for existing utility measures,
- whether they are complementary,
- whether they represent a fundamentally different construct.

10 Insufficient Psychometric Framing

The project repeatedly discusses:

- recoverability,
- inference,
- latent variables,
- uncertainty,

but provides little formal psychometric grounding.

Important missing concepts include:

- construct validity,
- convergent validity,
- discriminant validity,
- test-retest reliability,
- measurement invariance,

- inter-rater reliability,
- calibration drift,
- latent factor structure.

Without a stronger psychometric framework, the proposal risks appearing technologically sophisticated but scientifically under-specified.

11 Methodological Concerns

12 Circularity and LLM Leakage

The largest methodological risk is circularity.

If one LLM generates synthetic patients and another LLM infers utilities, then the inference model may exploit stylistic artifacts, shared priors, latent assumptions, recognizable and narrative structures, rather than genuinely recovering utilities. This could dramatically overestimate recoverability.

The proposal acknowledges this issue but underestimates its severity.

Potential mitigation strategies should include:

- cross-model evaluation,
- adversarial prompting,
- paraphrase perturbation,
- stylistic randomization,
- human-authored transcripts,
- blinded generation conditions.

13 Synthetic Patients May Be Unrealistically Coherent

LLMs tend to generate:

- internally coherent,
- semantically organized,
- psychologically interpretable narratives.

Real patients are often:

- contradictory,
- indirect,
- avoidant,

- distracted,
- inconsistent,
- culturally constrained,
- emotionally guarded.

This may make the inference problem artificially easy.

The proposal currently lacks:

- explicit noise models,
- inconsistency simulation,
- conversational derailment,
- memory limitations,
- emotional suppression models,
- linguistic variability models.

14 Identifiability May Be Much Weaker Than Assumed

The proposal assumes utilities are recoverable from narrative.

This may be true for:

- highly salient impairments,
- socially disruptive symptoms,
- behaviorally visible losses.

However, many utilities may be weakly identifiable.

For example:

- two patients may produce similar narratives for entirely different reasons,
- adaptation may suppress mention of high-impact impairments,
- cultural norms may alter expression,
- stoicism may conceal burden.

The proposal would benefit from a more explicit discussion of:

- identifiability limits,
- irreducible ambiguity,
- latent-variable nonuniqueness.

15 Lack of Formal Causal Structure

The proposal informally references symptoms, functional effects, utilities, and narratives, but does not specify causal relationships clearly.

For example:

- Do utilities influence topic salience?
- Does adaptation mediate symptom expression?
- Does personality affect linguistic style independently of utility?
- Are narratives generated from objective state or subjective burden?

Without a clearer causal model, inference risks becoming descriptive rather than explanatory.

16 Clinical and Translational Concerns

17 Real Patients May Resist Open-Ended Narrative Interviewing

The proposal assumes patients will willingly provide rich narratives.

In practice:

- some patients are terse,
- some dislike open-ended questions,
- some avoid emotional disclosure,
- some focus on biomedical details,
- some become fatigued quickly.

Narrative richness may vary dramatically across:

- age,
- culture,
- education,
- personality,
- clinical setting.

18 Clinical Workflow Burden Is Underexplored

Even if conversational elicitation works technically, important practical questions remain:

- How long are interviews?
- Who conducts them?
- How are transcripts reviewed?
- How is clinician time affected?
- How are conflicting inferences resolved?
- How are uncertainty estimates communicated?

The proposal currently focuses more on methodological novelty than operational feasibility.

19 Unclear Regulatory and Clinical Positioning

The proposal does not clearly specify whether the intended use case is:

- research,
- shared decision-making,
- clinical decision support,
- quality-of-life assessment,
- utility estimation for economic modeling,
- patient-centered interviewing augmentation.

This ambiguity may create skepticism among healthcare reviewers.

20 Ethical Concerns

21 Risk of Over-Inference

Inferring latent preferences from conversation raises important ethical issues.

Potential risks include:

- inferring values patients did not intend to reveal,
- demographic stereotyping,
- algorithmic overconfidence,
- false assumptions about patient priorities.

The proposal should more clearly distinguish exploratory inference from authoritative preference determination.

22 Privacy Concerns

Conversational transcripts may contain:

- highly sensitive information,
- family details,
- emotional disclosures,
- social vulnerabilities

The proposal should discuss:

- de-identification,
- local-only processing,
- transcript retention,
- consent procedures,
- governance models.

23 Technical Concerns

24 Human-Mediated Prompting Limits Scalability

The copy/paste architecture is elegant for reproducibility and cost control, but creates practical limitations:

- manual workflows,
- transcription errors,
- inconsistent prompting,
- poor scalability,
- limited automation.

This is acceptable for a research prototype but may constrain large-scale experimentation.

25 Browser-Based Architecture May Limit Advanced Modeling

A fully client-side architecture may eventually constrain:

- probabilistic inference,
- large embedding models,
- Bayesian sampling,
- transcript indexing,
- active-learning workflows.

The proposal should discuss:

- future migration paths,
- optional server-side extensions,
- interoperability with external analysis pipelines.

26 Missing Evaluation Strategy

The proposal describes many possible metrics but lacks:

- a rigorous experimental plan,
- baseline comparisons,
- statistical power considerations,
- predefined success criteria.

Critical missing comparisons include:

- conversational elicitation vs EQ-5D,
- conversational elicitation vs standard gamble,
- conversational elicitation vs clinician judgment,
- human raters vs LLM raters.

27 Recommendations

28 Clarify the Target Construct

The proposal should explicitly define:

- what "utility" means,
- whether conversationally inferred utilities are intended to approximate QALYs,
- how they relate to conventional utility frameworks.

29 Add Formal Psychometric Structure

The project would benefit substantially from:

- latent-variable modeling language,
- reliability analyses,
- construct-validity framing,
- measurement-error models.

30 Explicitly Model Failure Modes

The proposal should discuss:

- when conversational elicitation fails,
- which domains are unrecoverable,
- how uncertainty is quantified,
- ambiguity detection.

31 Separate Simulation and Inference More Aggressively

The proposal should adopt stronger anti-circularity safeguards:

- different models,
- paraphrasing layers,
- adversarial transcript transformations,
- human-authored validation corpora.

32 Emphasize Benchmarking Rather Than Clinical Replacement

The project is strongest as a methodological research platform, not a replacement for established utility elicitation methods. This framing would likely increase reviewer confidence.

33 Overall Assessment

This proposal is intellectually ambitious, methodologically creative, technically pragmatic, and potentially high-impact.

Its strongest contribution is likely to be a benchmark environment for studying conversational preference inference.

Its weakest areas involve:

- psychometric grounding,
- construct definition,
- validation strategy,
- causal specification,
- real-world clinical feasibility.

The project has substantial research value even if conversational utility recovery ultimately proves limited.

Indeed, demonstrating the limits of recoverability may itself be an important scientific contribution.

The proposal is therefore best viewed as an exploratory methodological research program, with strong potential for innovation, but also substantial uncertainty regarding identifiability, validity, and translational feasibility.